

Yurina Nakazato

Curriculum Vitae (last updated July 2025)

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Research Interests

Galaxy formation and evolution, cosmology, first stars and galaxies, cosmological simulations, multi-wavelength observations

Education

- Apr. 2023 - present **Ph.D. student, The University of Tokyo**, Japan
(Ph.D. expected, August 2025)
Thesis: Formation and Evolution of High-Redshift Galaxies Using Cosmological Simulations and Multi-Wavelength Observations
Advisor: Naoki Yoshida
- Apr. 2021 - Mar. 2023 **M.S. in Physics, The University of Tokyo**, Japan
Thesis: *Formation and evolution of star clusters and galaxies in the early Universe*
Advisor: Naoki Yoshida
President's Award (1/3000), Dean's Award (1/128)
- Apr. 2017 - Mar. 2021 **B.S. in Physics, The University of Tokyo**, Japan

Fellowships and Awards

- Feb. 2024 Outstanding Poster Award at the International Symposium on Quantum Electronics 2024
- Apr. 2023- present JSR fellowship
- Apr. 2023- present Japan Society for Promotion of Science (JSPS) Research Fellow, DC1
- Mar. 2023 The School of Science Encouragement Award
- Mar. 2023 The University of Tokyo President's Award
- Mar. 2023 The Excellence Award for Qualifying Exam
- Oct. 2021- present The International Graduate Program for Excellence in Earth-Space Science (IGPEES), the University of Tokyo
- Aug. 2021 Best Oral Presentation Award at 51th astronomical meeting for junior researchers

Leadership & Outreach

- Jun. 2023 Organizer of Get-Together Event for Women in STEM, the University of Tokyo
- Jul. 2022 Public Talk at Women in STEM, School of Science, the University of Tokyo
- Dec. 2019 Rikejo Initiative, the University of Tokyo
- Oct. 2019 Invited talk at Dow Chemical Company, Tokyo, Japan

Grants

- Apr. 2023-Mar.2026 Evolution of first galaxies with multi-wavelength observations and numerical simulations, 4.2M JPY (30K USD), JSPS Grant-in-Aid for Early-Career Scientists, No. 23KJ0728
- Apr. 2023-Mar.2026 Grand Aid from JSR fellowship, 3M JPY (21K USD)

Talks at Conferences and Workshops (Selected)

total: 25 (24 oral talks, 9 peer-reviewed talks)

- Sep. 2024 **ILR workshop 2024 @Osaka: From Galaxy Formation to Supermassive Black Holes**, Osaka, Japan
- Aug. 2024 **IAUS391**, Cape Town, South Africa
- Jul. 2024 **European Astronomical Society (EAS) Annual meeting 2024**, Padova, Italy
- Feb. 2024 **The International Symposium on Quantum Electronics 2024**, poster, The University of Tokyo, Japan
- Nov. 2023 **Resolving the Extragalactic Universe with ALMA & JWST**, Waseda Univ., Japan
- Jul. 2023 **Shedding new light on the first billion years of the Universe**, Marseille, France
- Sep. 2022 **The 9th East Asian Numerical Astrophysics Meeting (EANAM9)**, Okinawa, Japan
- Aug. 2022 **Star Formation in Different Environments (SFDE) 2022**, Quy Nhon, Vietnam
- Jul. 2022 **National Astronomy Meeting (NAM) 2022**, online, hosted by the University of Warwick
- Nov. 2021 **IAU Symposium 362: The predictive power of computational astrophysics as a discovery tool**, online
- Aug. 2021 **RESCEU Summer School 2021**, online, hosted by the University of Tokyo

Seminars (Selected)

- Dec. 2024 **One-day workshop on high-z cosmological simulations**, The University of Tokyo
- Jul. 2024 **Special seminar**, JSR corporation, Online
- Apr. 2024 **Special seminar**, Scuola Normale Superiore, Italy
- Dec. 2024 **Special seminar**, Universidad Autónoma de Madrid, Spain
- Dec. 2024 **Special seminar**, Centro de Astrobiología, (CAB, CSIC-INTA), Spain
- Aug. 2023 **One-day workshop on galaxies in the era of JWST/ALMA**, The University of Tokyo
- Mar. 2023 **Supersonic Project: Collaboration meeting**, UCLA, US
- Feb. 2023 **Special seminar**, UCLA, US

Coverage in Media

Hunting Faint Galaxies with JWST as a Test for the Dark Matter Model based on Williams et al. 2024
- *Bright galaxies put dark matter to the test*, UCLA

The core of the most distant galaxy protocluster observed by JWST & ALMA based on Hashimoto et al. 2023

- *The Strong Tag Team of the James Webb Space Telescope and ALMA Captures the Core of the Most Distant Galaxy Protocluster*, Kavli IPMU, Tsukuba Univ., Waseda Univ., Nagoya Univ.

Teaching & Professional Service

Teaching

Apr. 2021-Aug. 2021 Teaching Assistant of Fluid Mechanics, the University of Tokyo
Apr. 2019- Jan. 2024 Language Assistant, International Lounge, the University of Tokyo

Professional Service

Apr. 2025 Organizer, One-day workshop on for stellar-cluster simulation, The University of Tokyo, Tokyo, Japan
Dec. 2024 Organizer, One-day workshop on high-z cosmological simulations, The University of Tokyo, Tokyo, Japan
Aug. 2023 Organizer, The 53rd Summer School on Astronomy and Astrophysics, The University of Tokyo, Tokyo, Japan
Mar. 2023 Workshop Organizer, Astrophysics Workshop for Young Researchers, The University of Tokyo, Tokyo, Japan

Proposals

co-Investigator on ALMA proposals

4. 2023.1.00843.S, PI: Y.Tamura, 15.6h
Disentangling the ionization structure of HII regions in a density-bounded Ly α emitter in the Reionization Era
3. 2023.1.00193.S, PI: Y. Fudamoto, 17.3h
[CII] 158 μ m emission line and dust observation of the most distant known overdensity of galaxies at $z = 7.88$
2. 2023.1.00235.S, PI: Y. Fudamoto, 10.0 h
The ALMA-JWST synergy: [OIII]88 μ m and [CII]158 μ m emission line observations of $z = 9.51$ galaxy identified by JWST
1. 2023.1.01362.S, PI: Y. Tamura, 25.9 h
Pandora's ELPIS: The Emission-Line Protocluster Imaging Survey of the furthest overdensity beyond Pandora's Cluster

Research & industry experience

Apr. 2024 - Jul. 2024 **Visiting Student at Scuola Normale Superiore di Pisa**
- Host researcher is Prof. Andrea Ferrara.

Nov. 2023 - Dec. 2023 **Visiting Student at Universidad Autónoma de Madrid**
- Host researcher is Prof. Daniel Ceverino.
- Worked on formation and evolution of clumpy galaxies by using FirstLight simulation

Feb. 2023 - Mar. 2023 **Visiting Student at UCLA**
- Host researcher is Prof. Smadar Naoz.
- Worked on Supersonic Project [link]

Jun. 2020 - Mar. 2021 **Study and Visit Abroad Program**
- Funded by the faculty of science, the University of Tokyo
- Online research internship at Naoz lab, UCLA

Dec. 2019 - Jan. 2020 **Online Language Exchange Program - UTokyo & TUM-**
- Participated in an online international exchange program with students from Technical University of Munich

Jun. 2019 - Sep. 2019 **UTokyo Global Internship Program**
- Funded by DAIKIN, a company leading air conditioning and refrigeration.

- Two-month workplace training at DAIKIN Japan and two-week research internship at DAIKIN Europe in Belgium. Business proposal for food loss and integrated solution of air conditioning and refrigeration.
- Aug. 2019 **Summer School of Particle Physics and Nuclear Physics**
 - Funded by the High Energy Accelerator Research Organization.
 - Experiment of measuring muon decay time and observing Lamor Precession and Single-Spin Asymmetries. Made the final presentation and poster session.
- Aug. 2019 **Nanotechnology Platform Student Training Program**
 - Funded by National Institute for Materials Science.
 - Five-day research program at Spring-8, the world's largest third-generation synchrotron radiation facility. Conducted X-ray photoelectron spectroscopy (XPS) experiments and data analysis. Made a presentation at the University of Tokyo in September.
- Feb. 2019- Mar. 2019 **Undergraduate Research Assistant, TOMODACHI STEM Program**
 - Funded by U.S.-Japan Council.
 - Five-week science & engineering research internship at Gerts lab, Rice University, Houston. Researched and analyzed the heavy iron collision data of STAR experiment conducted at BNL.
 - Final week study tour to Washington, DC including site visits to the Society for the Promotion of Science, JAXA, U.S.-Japan Council, and Women in STEM Workshop at Lehigh University.
- Jul. 2014- Aug. 2014 **Okinawa Global Leaders Program**
 - Funded by Okinawa Prefectural Board of Education.
 - Three-week program to introduce students to key concepts in intercultural communication and global leadership.

Active collaborations

- Supersonic Project
- FirstLight simulations
- Reionization and the ISM/Stellar Origins with JWST and ALMA (RIOJA, JWST GO1840)

Competencies

Scientific programming : Python, C++, Fortran, Latex
 Simulation software : SKIRT, AREPO, ART
 Astrophysics tool : CLOUDY, SKIRT, Astrodendro, A-SLOTH

References

Prof. Naoki Yoshida (current supervisor), The University of Tokyo, naoki.yoshida@ipmu.jp
 Prof. Andrea Ferrara, Scuola Normale Superiore, andrea.ferrara@sns.it
 Prof. Smadar Naoz, University of California, Los Angeles, snaoz@astro.ucla.edu